

Course Code MAT2005	Transform Techniques and Difference Equations												Course Type	LT		
													Credits	3		
													Version	1.0		
Prerequisite:	MAT1003 - Calculus															
Course Objectives: • This course is designed to give a comprehensive coverage of transform techniques and difference equations. • To represent the periodic/non-periodic function using Fourier series/Fourier transform. • To solve differential equations using Laplace transform. • To study applications of Laplace transform in Signal & System and Network analysis. • To study the solution discrete problem in terms of difference equations. • To find solution of difference equations using Z-transform.																
Course Outcomes (CO): Students will be able to CO1: Apply and analyze the concept of FS and FT in Signal and Systems. KL3, KL4 CO2: Apply Laplace transform to solve an ordinary differential equation. (KL4) CO3: Understand the concept of Z-Transform. KL2, KL3 CO4: Understand the difference equation and solution of difference equation using Z-Transform. KL3, KL4																
Correlation of COs with Pos																
CO\PO	CKL	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
PKL		3	5	6	5	6	3	3	3	NA	M	3	M	3	3	
CO1	3	3	3	3	2	2	0	0	2	0	2	2	2			
CO2	4	3	3	3	3	2	0	0	2	0	2	2	2			
CO3	3	3	3	3	2	3	2	2	0	2	2	2	2			
CO4	4	3	3	2	3	3	0	0	0	2	2	2	2			
COs	Topics to be discussed														Lectures	
CO1	Fourier series Introduction to Fourier series - Euler formulas for the Fourier coefficients - Dirichlet's conditions - Fourier convergence Theorem (Statement) - Fourier series over a shifted interval - Half range Fourier series - Parseval's identity - Frequency and amplitude spectra of a function - computation of harmonics.														7	
CO1	Fourier Transform Fourier Integral Theorem (statement only) - Fourier Sine and Cosine Integral Theorem (statement only), Fourier Transform - Fourier Cosine & Sine Transforms of elementary functions. Properties of Fourier Transform: Linearity, Shifting, Change of scale, Fourier Transform of Derivatives, Convolution Theorem (statement only), Inverse of Fourier Transform, Application of Fourier Transform in Real Life.														6	
CO2	Laplace Transform Introduction to Laplace transform - conditions for existence - Laplace transforms of elementary functions, periodic functions, unit step and unit impulse functions - properties - Introduction to Inverse Laplace transform - Inverse Laplace transform by partial fraction method and convolution theorem - solution of differential equations using Laplace Transforms.														7	
CO3	Z-Transform Definition of Z-transform - Z-transform of discrete sequences and functions - Relation between Z transform and Laplace transform - Definition of inverse Z transform - Inverse Z transform by partial fraction and convolution methods.														5	

CO4	Difference Equations Introduction to Difference equation - order and linear - Fibonacci sequence - solution of linear difference equations with constant coefficients - complementary functions - particular integrals by the method of undetermined coefficients - solution of simple difference equations using Z-transforms, Application of Difference equations.	5
Guest Lectures by experts on contemporary topics		2
Total Lectures: (1 Lecture = 1.5 hrs.)		32
Text Books: 1. Erwin Kreyszig, Advanced Engineering Mathematics, 10th Edition, Wiley India Pvt. Ltd., New Delhi, 2015. 2. B. S. Grewal, Higher Engineering Mathematics, 43rd Edition, Khanna Publishers, India, 2014. 3. B. V. Ramana, Higher Engineering Mathematics-McGraw-Hill Education, 2006.		
Reference books: 1. Peter V. O' Neil, Advanced Engineering Mathematics, 7th edition, Cengage Learning, 2012. 2. William E. Boyce, Richard C. DiPrima, Elementary differential equations and boundary value problems, 10th edition, John Wiley & Sons, Inc., 2012. 3. Michael E. Greenberg, Advanced Engineering Mathematics, 2nd edition, Pearson, 2006.		
Recommendation by the Board of Studies on		15 March 2024
Approval by Academic council on:		23.05.2024
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